APTOS DR Model Training Notebook — Summary

Model Architecture and Training Mechanism

The predictive model is built on a ResNet-18 architecture that was pre-trained on ImageNet. The network is fully fine-tuned without frozen layers, and the final layer has been replaced to output 5 logits corresponding to the Diabetic Retinopathy (DR) severity grades 0 through 4.

- **Input parameters**: Images are scaled to 224x224 RGB pixels and normalized using standard ImageNet mean and standard deviation values.
- **Augmentation**: During the training phase, the model applies random horizontal flipping and random rotations of up to ±10°.
- **Loss and Optimization**: The pipeline uses a cross-entropy loss function, heavily weighted by inverse class frequency to manage class imbalance. The optimizer is AdamW with a learning rate of 1e-4 and a weight decay of 1e-4.
- **Training Loop**: The model is configured to run for a maximum of 20 epochs with an early stopping patience of 5 epochs. Checkpoints are saved when the validation macro-F1 score improves, while the validation Quadratic Weighted Kappa (QWK) is tracked as a secondary metric for ordinal grading.

Data Engineering and Preprocessing Pipeline

The model training notebook does not independently fetch data; it relies on a validated handoff manifest (split_manifest.csv) and a directory of preprocessed images (data/processed_224_v1/) produced by the separate data engineering workflow.

- **Preprocessing applied**: All eligible original images were converted to RGB, resized using aspect-preserving LANCZOS interpolation, and padded with black borders to fit the 224x224 target resolution.
- **Integrity gates**: Before training begins, the notebook verifies matching IDs, prevents duplicates, checks split labels, and confirms the 224x224 resolution. It will raise a hard error if any condition fails rather than proceeding.

Dataset Splits and Class Distribution

The data engineering pipeline successfully processed 3,662 original images. Exactly 62 records were provisionally excluded because identical-image groups exhibited conflicting severity grades. The remaining 3,600 images were partitioned into training (roughly 70%), validation (15%), and testing (15%) splits.

Grade	Train	Validation	Test	Excluded
0	1258	270	273	4
1	253	54	53	10

Grade	Train	Validation	Test	Excluded
2	684	148	144	23
3	127	26	27	13
4	199	40	44	12

The table above details the final class distribution of the 3,600 partitioned images and the 62 excluded conflicting records across the five diagnostic grades. The test split is explicitly locked out of model selection and cannot be evaluated unless the `CONFIRM_REVIEWED_FROZEN_SPLIT` flag is manually engaged.

Provisional Results and Model Metrics

The notebook currently displays an execution log of 9 epochs, with the best model checkpoint occurring at epoch 4.

Metric	Recorded Value
Validation accuracy	82.2%
Validation macro-F1	0.661
Validation QWK	0.875
Test evaluation	Skipped

These results show strong performance on grade 0 (0.97 F1) but weak performance on severe grades, with grade 3 logging a 0.45 F1 and grade 4 logging a 0.49 F1. The model frequently misclassified these severe instances as grade 2.

Critical Data Conflict: The opening cell of the notebook explicitly states that no model has been trained, and no performance results are available, directly contradicting the displayed outputs. These metrics are likely placeholder remnants from a previous run and must be treated as completely unverified until the notebook is re-run cleanly.

Limitations and Final Verification

This pipeline represents a reproducible baseline rather than a clinically meaningful diagnostic tool. The data split remains strictly provisional because 4,096 near-duplicate image candidate pairs have yet to be manually reviewed and resolved. Because patient-level independence across the training and validation sets has not been confirmed, there is a risk that leaked near-duplicates are artificially inflating validation metrics like the 82.2% accuracy score. No current metrics should be quoted or presented as final evidence until the split is frozen, and the data inconsistencies are fully resolved.